

React HandsOn 10:-

THEORY

Define JSX

JSX (JavaScript XML) is a syntax extension for JavaScript that allows developers to write HTML-like code inside JavaScript. It makes React code easier to read and write.

Example

```
const element = <h1>Hello World!</h1>;
```

Instead of writing:

```
const element = React.createElement("h1", null, "Hello World!");
```

Features

- Looks like HTML.
- Easier to understand.
- Can contain JavaScript expressions.
- Used to create React elements.

Explain about ECMA Script

ECMAScript (ES) is the standard on which JavaScript is based.

ES6 (ECMAScript 2015) introduced many new features used in React.

Important ES6 Features

a) let

Block-scoped variable.

```
let age = 20;
```

```
age = 21;
```

b) const

Constant variable.

```
const PI = 3.14;
```

Cannot be reassigned.

c) Arrow Functions

```
const greet = () => {  
    console.log("Hello");  
};
```

Shorter syntax than normal functions.

d) Classes

```
class Student{  
    constructor(name){  
        this.name = name;  
    }  
}
```

Used to create objects.

e) Modules

```
export default App;  
  
import App from './App';
```

Allows code reuse.

f) Destructuring

```
const person = {  
    name: "John",  
    age: 22  
};  
  
const {name, age} = person;
```

g) Spread Operator

```
const arr1 = [1,2];
```

```
const arr2 = [...arr1,3,4];
```

Explain React.createElement()

Every JSX code is converted into React.createElement().

Syntax

```
React.createElement(type, props, children);
```

Example

Without JSX

```
const element = React.createElement(  
  "h1",  
  null,  
  "Welcome"  
);
```

With JSX

```
const element = <h1>Welcome</h1>;
```

Both create the same React element.

Parameters

- type → HTML tag or React component
- props → Attributes
- children → Content inside the element

Example

```
React.createElement(  
  "h2",  
  {style:{color:"blue"}},  
  "Hello"  
);
```

Equivalent JSX

```
<h2 style={{color:"blue"}}>Hello</h2>
```

Explain how to create React nodes with JSX

React nodes are created simply by writing JSX.

Example

```
const element = <h1>Hello React</h1>;
```

Multiple Elements

```
const element = (  
  <div>  
    <h1>Student</h1>  
    <p>React Learning</p>  
  </div>  
);
```

The <div> is called the parent element.

Using Variables

```
const name = "Arpita";  
const element = (  
  <h1>Hello {name}</h1>  
);
```

Output

Hello Arpita

Define how to render JSX to DOM

Rendering means displaying React elements on the webpage.

In React 18:

index.js

```
import React from "react";
```

```
import ReactDOM from "react-dom/client";
```

```
import App from "./App";
```

```
const root = ReactDOM.createRoot(document.getElementById("root"));
```

```
root.render(<App />);
```

App.js

```
function App(){
```

```
  return(
```

```
    <h1>Hello React</h1>
```

```
  );
```

```
}
```

```
export default App;
```

index.html

```
<div id="root"></div>
```

React inserts the JSX inside the root element.

Explain how to use JavaScript expressions in JSX.

JavaScript expressions are written inside curly braces {}.

Example 1

```
const name = "Arpita";
```

```
function App(){
```

```
  return(
```

```
    <h1>Hello {name}</h1>
```

```
  );
```

```
}
```

Output

Hello Arpita

Example 2

```
const a = 10;
```

```
const b = 20;
```

```
<h2>{a + b}</h2>
```

Output

30

Example 3

```
const marks = 90;
```

```
<h1>{marks >= 50 ? "Pass" : "Fail"}</h1>
```

Output

Pass

Example 4

```
const today = new Date();
```

```
<p>{today.toDateString()}</p>
```

Explain how to use inline CSS in JSX

In React, CSS is written as a JavaScript object.

Syntax

```
style={{property: "value"}}
```

Example

```
<h1 style={{color:"red"}}>
```

 Hello

```
</h1>
```

Multiple Properties

```
<h1
```

```
  style={{
```

```
    color:"blue",
```

```
    backgroundColor:"yellow",
    fontSize:"30px"
  }}>
Welcome
</h1>
Using a Style Object
const myStyle = {
  color:"green",
  backgroundColor:"lightgray",
  padding:"20px",
  fontSize:"25px"
};
function App(){
  return(
    <h1 style={myStyle}>
      React Styling
    </h1>
  );
}
```

HANDS-ON LAB

Office Space, at Affordable Range



Name: DBS

Rent: Rs. 50000

Address: Chennai



Name: Regus

Rent: Rs. 75000



Name: Regus

Rent: Rs. 75000

Address: Bangalore



Name: WeWork

Rent: Rs. 45000

Address: Hyderabad

